# Do System One Models Make Better Rerankers?

[TODO: optional subtitle, e.g. “Decision Fine-Tuning versus Reranker Fine-Tuning on Six Foundation Models”]

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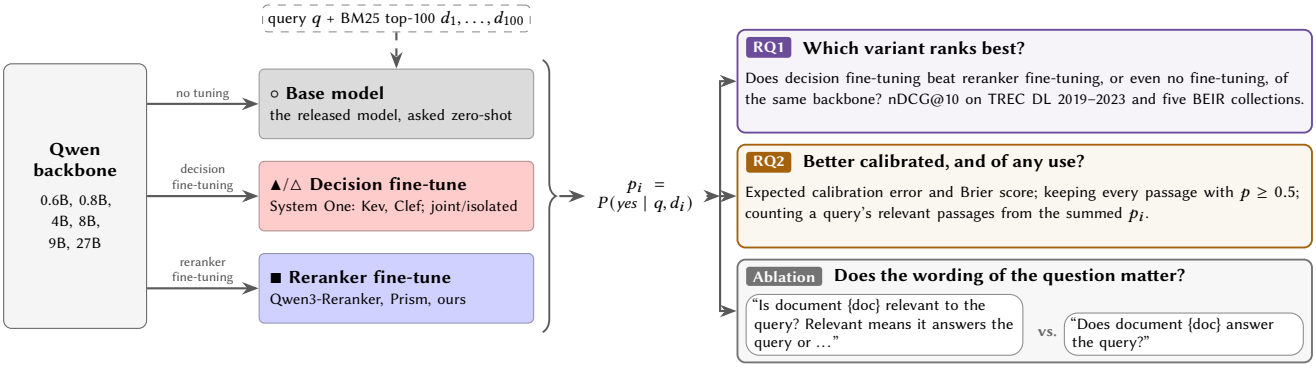



**Figure 1: What we ask. Each Qwen backbone is compared in three variants that all return a probability of relevance for a query and a BM25 candidate: the base model asked zero-shot, its decision fine-tunes (System One models such as Kev and Clef) and its reranker fine-tunes. RQ1 asks which variant ranks best, RQ2 whether the decision models are at least better calibrated and whether that is of any use. An ablation asks how much both answers depend on the wording of the question put to the model.**

## Abstract

System One models, or decision models, answer a question about an input with a probability instead of generated text. A month after the first of them was released, they are marketed for their calibrated probabilities and already used as rerankers. We ask whether the decision fine-tuning behind them makes a better reranker than the established alternative, plain reranker fine-tuning of the same model. For six Qwen backbones from 0.6B to 27B parameters and ten TREC DL and BEIR collections, we compare the zero-shot base model with its decision fine-tunes and its reranker fine-tunes, two of which we train ourselves on MS MARCO. Up to 9B, reranker fine-tuning wins on every backbone, by 5 to 29 points nDCG@10, also on collections that no fine-tune was trained on, while decision fine-tuning does no better than simply asking the base model. At 27B, the best decision fine-tune and our reranker fine-tune rank equally well. The calibration that decision models are advertised for does not rescue them either. Whether they are better calibrated than reranker fine-tunes flips with the wording of the relevance question; at 9B, our reranker fine-tune, trained with a plain yes/no loss, is as well calibrated as the best decision model; and up to 9B, a reranker fine-tune keeps the better passages at a fixed threshold. Only at 27B does decision fine-tuning have an edge, and it lies in calibration, not in ranking. As released so far, decision fine-tuning gives no reason to replace a reranker.

## Keywords

Reranking, Calibration, System One Models, Decision Models, Fine-Tuning, Large Language Models

## 1 Introduction

Cross-encoders are a well-established way to rerank the candidates of a first-stage retriever [19, 20], and open large language models fine-tuned for ranking are the current state of the art [17, 38, 39]. A recent alternative are *System One models*, also called decision models. They answer a question about an input with a probability for each possible answer instead of generating text, in the spirit of fast, intuitive thinking [14, 33]. TypeSafe AI released the first of them, Jev, on 15 September 2026, trained with a method it calls RLCD (reinforcement learning for calibrated decisions) and has not published [33]. Within weeks, Jev was reported to be a competitive reranker [13, 27] and was evaluated or integrated in retrieval systems [11, 15, 34]. Open decision models followed: Kev by the end of September [22] and Cloudflare's Clef on 1 October [4]. They are fine-tuned from Qwen backbones, the same backbones that dedicated rerankers such as Qwen3-Reranker [39] and Prism [38] are built on.

The main selling point of decision models is that their probabilities are calibrated. Reranking, however, depends on the order of the scores, not on their values: any monotone transformation of the probabilities leaves the ranking unchanged, so calibration alone does not make a better reranker. The reports so far compare Jev with rerankers built on other, mostly smaller models. They cannot

tell how much of its effectiveness is due to decision fine-tuning and how much to the capability of the underlying language model. We argue that a fair evaluation of decision models as rerankers has to control for the backbone and compare against a reranker fine-tune of the same backbone. It should also test whether the advertised calibration holds for relevance, and whether it is of any use once the ranking is worse. We ask:

**RQ1** Do decision models rerank better than their base models or than reranker fine-tunes of the same backbone?

**RQ2** Are their relevance probabilities better calibrated than either, and is that calibration of any use?

Figure 1 shows the design. For six Qwen backbones from 0.6B to 27B parameters, we compare three variants of the same model: the base model asked zero-shot, its decision fine-tunes (Kev and Clef, in two modes) and its reranker fine-tunes. No reranker fine-tune of the 27B backbone has been released, so we train one ourselves, and a second one at 9B as a control, both on MS MARCO alone with standard losses. All variants rerank the same BM25 candidates of TREC DL 2019–2023 and five BEIR collections, and all return a probability of relevance per passage. Every decision and base model is asked the same, plain question: does the passage answer the query? An ablation replaces it with the longer question of S1Rank [27], the first public evaluation of Jev as a reranker.

Both answers are negative.

- *Reranker fine-tuning wins.* Up to 9B, the reranker fine-tune ranks better than the decision fine-tune on every backbone, by 5 to 29 points nDCG@10 on average, significantly on 48 of 60 pairs of backbone and collection and never significantly worse. This holds on collections that no fine-tune was trained on.
- *Decision fine-tuning adds nothing up to 9B.* It stays within −2.5 and +0.7 points of simply asking the base model, while reranker fine-tuning adds 6 to 30 points. Only Kev-27B clearly improves on its base model, and our own 27B reranker fine-tune, trained with a public recipe, ranks as well as it does.
- *Calibration does not rescue decision models.* With our question they are better calibrated than the published reranker fine-tunes from 4B on; with S1Rank's question they are not. Our reranker fine-tune of Qwen3.5-9B, trained with a plain yes/no loss, is as well calibrated as the best decision model of its size. And up to 9B, a reranker fine-tune keeps the better set of passages at a fixed threshold. Only at 27B is the decision fine-tune better calibrated than our reranker fine-tune.

In short: if you need a reranker, fine-tune a reranker. Decision models may have their uses, but as released so far, reranking is not one where they beat a model trained for it. Code, raw model outputs (a probability per query and passage) and all results are public.[1]

[1] [TODO: repository link (code release: branch `feature/release-20261012`)]

## 2 Related Work

*LLM-based rerankers.* Neural rerankers are trained to score query–passage pairs from relevance labels, first with encoder or sequence-to-sequence models [19, 20] and more recently with open large language models [17, 38, 39]. Qwen3-Reranker and Prism read relevance from the probability of "yes" against "no" after a prompt that asks whether the document meets the query, which makes their output directly comparable to that of a decision model. Other approaches rerank the candidates jointly, either by generating a ranked list [24, 31] or by letting passages attend to each other, as in the Set-Encoder [28]. Large language models can also rank without any reranking-specific fine-tuning, through relevance prompting [26, 40], but their effectiveness is sensitive to how the prompt is worded [30]. Abdallah et al. [1] compare many of these rerankers on the candidates we use.

*Decision models in retrieval.* Besides Jev [33], Kev [21, 22], Clef [4] and Laya [5] are open System One models. The evidence on decision models for retrieval is so far limited to reports that are not peer-reviewed: Jev reranking BM25 candidates [13, 15, 27], as a reranker in production search [11, 34], and Laya as a relevance gate before recall [29]. These reports compare Jev with existing rerankers, which are built on other models. None compares decision fine-tuning with reranker fine-tuning of the same backbone, and none asks whether the calibration of the probabilities pays off.

*Calibration of relevance scores.* Neural networks tend to be overconfident, and temperature scaling after training is a simple remedy [12]. Neural rankers are no exception [23]. Common ranking losses are invariant to shifting the scores of a query, so rankers trained with them can order passages well without being calibrated [23, 36]; scores can be calibrated during or after training [12, 37]. Calibrated probabilities matter wherever a score is read on an absolute scale, for example to cut off a ranked list [2]. This is the use we test in RQ2.

## 3 Decision Models as Rerankers

A decision model takes a structured input, the *state*, together with one or more questions about it, and returns a probability for each possible answer. We use binary questions only, so every question yields one number, the probability of "yes". Reranking follows directly: for a query $q$ with candidates $d_1, \ldots, d_k$ from a first-stage retriever, we ask about each passage $d_i$ whether it answers $q$ and order the passages by the returned probabilities $p_i$.

For the state, we consider two modes. In *isolated* mode, the state holds the query and a single passage, so that $p_i = P(yes \mid q, d_i)$. This is a pointwise reranker, like a cross-encoder. In *joint* mode, the state holds the query and all $k$ candidates in first-stage order, and the $k$ questions are asked about this one state, so that $p_i = P(yes \mid q, d_1, \ldots, d_k)$ can depend on the other candidates. Joint mode shares its input with listwise rerankers but returns a score per passage instead of a permutation, like the Set-Encoder [28]. Both modes are pointwise in what they output, and both are the modes of S1Rank [27].

The two alternatives have the same interface. A reranker fine-tune such as Qwen3-Reranker reads $p_i$ from the probability of "yes" against "no" after a prompt that holds $q$ and $d_i$. A base model, asked

the decision models' question in its chat format, gives $p_i$ as the probability of "yes" as its first answer token. All three variants of a backbone therefore produce a probability of relevance for every query and passage, which we can rank by (RQ1) and read on an absolute scale (RQ2).

## 4 Experimental Setup

*Backbones and variants.* We study six Qwen backbones: Qwen3-0.6B and 8B, Qwen3.5-0.8B, 4B and 9B, and Qwen3.8-27B. Each is evaluated in up to three variants (Figure 1). The *base model* is the released model, asked the question in its chat format with thinking off and scored by the probability of "yes" as the first answer token. The *decision fine-tunes* are Kev [21, 22] (LoRA on the pretrained checkpoints up to 9B, a full fine-tune of the post-trained 27B model) and Clef-Flash (9B) and Clef (27B) [4], each run in isolated and joint mode (Sect. 3). We call them decision fine-tunes because they fine-tune an open model to give Jev's kind of output; they are not reproductions of RLCD, which is unpublished [21]. The *reranker fine-tunes* are Qwen3-Reranker [39], built on the same pretrained checkpoints as Kev, and Prism [38], both scored by the probability of "yes" against "no".

*Our reranker fine-tunes.* No reranker fine-tune of Qwen3.8-27B has been released, so we train one ourselves (*Ours-27B*), from exactly the checkpoint that Kev-27B and Clef start from. As a control, we train a second one with the same recipe on Qwen3.5-9B (*Ours-9B*), where it can be compared with Prism-9B. The recipe is public and deliberately plain: LoRA ($r = 32$) in Qwen3-Reranker's prompt format, one epoch over 100k MS MARCO training queries [18], each with one relevant passage and 15 hard negatives (eight from a dense retriever, seven from BM25), and a ranking loss (InfoNCE over the group) plus a yes/no classification loss (binary cross-entropy of every passage's score). The second term gives the model a meaningful $P(\textit{yes})$. We removed training queries that match a test or development topic and chose the checkpoint on 200 MS MARCO dev queries by a rule fixed in advance. No decision data and no hyperparameter search are involved. Qwen3-Reranker, Prism and our fine-tunes saw MS MARCO in training; Kev's training data contain neither MS MARCO nor BEIR [21].

*The question.* Every decision and base model is asked one question per passage: "Does document `{doc}` answer the `query`?" It matches the relevance labels we evaluate against (on TREC DL, a passage counts as relevant if it answers the query, from grade 2). We fixed this question before running the experiments, but did not choose it on development data, and we report every result with it. The only other wording we ran is the question of S1Rank [27]: "Is document `{doc}` relevant to the `query`? Relevant means it answers the query or contains information that directly addresses it." It is the subject of the ablation in Sect. 7. Reranker fine-tunes ask no question of this kind; they keep their own prompt.

*Collections and candidates.* We use TREC DL 2019–2023 [6–10] on the MS MARCO v1 and v2 passages [18] and five BEIR collections [32]: TREC-COVID, NFCorpus, SciFact, Touché and DBPedia. None of the fine-tunes was trained on TREC-COVID, Touché or DBPedia, which form the *zero-shot subset* (ZS). Every system reranks the BM25 top-100 of Pyserini [16]; these are the candidates of S1Rank [27] and of earlier comparisons [1, 24, 31], whose published numbers we use where we did not rerun a system.

*Other systems.* For context, we report BM25, Jev 1.13.0 (closed, in both modes), Laya [5] (isolated only, as its context is too short for a joint state), monoT5-3B [20], bge-reranker-v2-m3 [3], the Set-Encoder [28], RankZephyr [24] (run by us on TREC DL) and RankGPT-4 [31] (published numbers only).

*Ranking.* We measure nDCG@10 and test differences with paired $t$-tests over queries, Holm-corrected across the ten collections ($p < 0.05$). Per backbone, we test the reranker fine-tune against the decision fine-tune with the best average (in its better mode), that decision fine-tune against the base model, and joint against isolated mode.

*Calibration and its uses.* We compute the expected calibration error (ECE, 15 equal-width bins) [12] and the Brier score (the mean squared error of the probability against the binary label, which also rewards separating relevant from non-relevant passages) on the judged query–passage pairs of TREC DL 2019–2023, the only collections that also judge many non-relevant passages; a passage counts as relevant from grade 2. We test two uses of the probabilities. A *fixed cutoff* keeps every candidate with $p \geq 0.5$; we report the $F_1$ of this set (unjudged passages count as non-relevant), its size, and its *regret*: the $F_1$ at the best threshold of each collection, chosen afterwards, minus the $F_1$ at 0.5. The *count* test sums a query's probabilities over its judged passages, which estimates how many of them are relevant, and reports the mean absolute error of this estimate in passages; always guessing the median count is the trivial baseline. Reranker fine-tunes are also calibrated after the fact, with temperature and Platt scaling of their logit fitted on 1,000 MS MARCO dev queries, never on a test collection.

*Sanity checks.* Our BM25, monoT5-3B and RankZephyr runs reproduce published numbers [1, 25] within 0.2 points nDCG@10. Jev in joint mode, asked S1Rank's question, reproduces S1Rank's numbers [27] within 0.5 points on four of five collections (1.5 below on TREC-COVID).

## 5 RQ1: Decision vs. Reranker Fine-Tuning

*Up to 9B, reranker fine-tuning wins on every backbone.* On all five backbones from 0.6B to 9B, the reranker fine-tune ranks better than the decision fine-tune in either mode (Table 1). Averaged over the ten collections, its margin over the better decision mode ranges from 5.2 points nDCG@10 at Qwen3.5-9B (Prism-9B 59.0 vs. Clef-Flash 53.8) to 28.8 points at Qwen3-0.6B (56.1 vs. 27.3). The difference is significant on 48 of the 60 pairs of reranker fine-tune and collection, and the decision fine-tune is never significantly better ($^*$). The margin is not an artefact of evaluating on MS MARCO. On the zero-shot subset, which no fine-tune was trained on, it ranges from 5.4 to 23.9 points. Nor does it depend on the undisclosed training recipes of the published rerankers: Ours-9B, trained on MS MARCO alone with standard losses, matches Prism-9B on average (59.4 vs. 59.0) and exceeds it on the zero-shot subset (56.1 vs. 53.7).

*Decision fine-tuning does no better than asking the base model.* Up to 9B, the better mode of a decision fine-tune stays within −2.5 and +0.7 points of its zero-shot base model, with significant gains on at

**Table 1: Base model, decision fine-tunes and reranker fine-tunes of the same backbone (nDCG@10, ×100, BM25 top-100). Base and decision models are asked whether the passage answers the query. Avg: mean over all ten collections; ZS: mean over the collections without a training split in any fine-tune's data (TREC-COVID, Touché, DBPedia). Per backbone and column: best in bold, second best underlined. Training: ○ base model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune (rows shaded grey, red, blue); *Ours*: our reranker fine-tunes. Paired $t$-test, Holm-corrected over the ten collections, $p < 0.05$: $^*$ reranker fine-tune vs. the decision fine-tune with the best Avg, $^\dagger$ that decision fine-tune vs. the base model, $^\ddagger$ joint vs. isolated.**

| | | | TREC DL | | | | | BEIR | | | | | Mean | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | System | Mode | DL19 | DL20 | DL21 | DL22 | DL23 | COVID | NFC | SciF | Touché | DBP | Avg | ZS |
| *Qwen3-0.6B* | | | | | | | | | | | | | | |
| ○ | Qwen3-0.6B | | 43.3 | 32.7 | 37.7 | 22.1 | 17.9 | 50.9 | 15.1 | 14.2 | 16.8 | 15.3 | 26.6 | 27.7 |
| ▲ | Kev-0.6B | joint | 23.3$^\ddagger$ | 15.1$^\ddagger$ | 20.4$^\ddagger$ | 12.8$^\ddagger$ | 12.1$^\ddagger$ | 36.8$^\ddagger$ | 12.4$^\ddagger$ | 4.5$^\ddagger$ | 5.2$^\ddagger$ | 8.4$^\ddagger$ | 15.1 | 16.8 |
| △ | Kev-0.6B | isolated | 35.3 | 30.5 | 40.1 | 22.8 | 19.0 | 55.9 | 16.5 | 15.5 | 14.1 | 22.9$^\dagger$ | 27.3 | 31.0 |
| ■ | Qwen3-Reranker-0.6B | | **69.5**$^*$ | **64.0**$^*$ | **65.2**$^*$ | **46.4**$^*$ | **38.8**$^*$ | **86.2**$^*$ | **36.4**$^*$ | **76.6**$^*$ | **35.7**$^*$ | **42.7**$^*$ | **56.1** | **54.9** |
| *Qwen3-8B* | | | | | | | | | | | | | | |
| ○ | Qwen3-8B | | 67.3 | 63.7 | 62.0 | 38.3 | 39.0 | 77.6 | 32.7 | 73.3 | 25.5 | 31.6 | 51.1 | 44.9 |
| ▲ | Kev-8B | joint | 62.7 | 55.2 | 59.1 | 35.5 | 34.3 | 53.1$^\ddagger$ | 20.4$^\ddagger$ | 17.9$^\ddagger$ | 12.3$^\ddagger$ | 36.2$^\ddagger$ | 38.7 | 33.9 |
| △ | Kev-8B | isolated | 69.0 | 63.1 | 61.9 | 38.6 | 36.4 | 80.6 | 33.8 | 69.8 | 25.2 | 39.0$^\dagger$ | 51.7 | 48.3 |
| ■ | Qwen3-Reranker-8B | | **73.2** | **66.8** | **67.0**$^*$ | **51.2**$^*$ | **45.2**$^*$ | **84.0** | **37.3**$^*$ | **79.9**$^*$ | **43.1**$^*$ | **46.2**$^*$ | **59.4** | **57.8** |
| *Qwen3.5-0.8B* | | | | | | | | | | | | | | |
| ○ | Qwen3.5-0.8B | | 46.0 | 41.2 | 52.1 | 28.1 | 27.9 | 59.2 | 18.3 | 27.0 | 16.4 | 22.3 | 33.9 | 32.6 |
| ▲ | Kev-0.8B | joint | 37.9 | 33.8 | 41.0 | 21.3 | 20.0 | 46.2$^\ddagger$ | 20.3 | 30.0 | 12.0 | 16.4$^\ddagger$ | 27.9 | 24.9 |
| △ | Kev-0.8B | isolated | 41.8 | 38.4 | 47.9 | 24.7 | 23.9 | 68.1$^\dagger$ | 19.2 | 31.5 | 16.5 | 29.9$^\dagger$ | 34.2 | 38.2 |
| ■ | Prism-0.8B | | **72.3**$^*$ | **67.7**$^*$ | **66.3**$^*$ | **47.0**$^*$ | **43.7**$^*$ | **84.2**$^*$ | **36.1**$^*$ | **76.3**$^*$ | **35.6**$^*$ | **41.7**$^*$ | **57.1** | **53.8** |
| *Qwen3.5-4B* | | | | | | | | | | | | | | |
| ○ | Qwen3.5-4B | | 67.9 | 64.0 | 65.9 | 42.3 | 40.4 | 78.0 | 33.3 | 74.1 | 25.0 | 33.0 | 52.4 | 45.3 |
| ▲ | Kev-4B | joint | 62.5 | 63.2 | 60.8 | 37.0 | 36.3 | 75.3 | 29.3$^\ddagger$ | 58.4$^\ddagger$ | 19.9 | 39.8 | 48.2 | 45.0 |
| △ | Kev-4B | isolated | 62.7 | 62.9 | 60.9 | 37.9 | 34.5$^\dagger$ | 78.0 | 32.0 | 67.7$^\dagger$ | 23.4 | 39.0$^\dagger$ | 49.9 | 46.8 |
| ■ | Prism-4B | | **74.4**$^*$ | **68.9**$^*$ | **70.3**$^*$ | **50.1**$^*$ | **46.2**$^*$ | **84.6**$^*$ | **37.4**$^*$ | **76.9**$^*$ | **34.3**$^*$ | **43.7**$^*$ | **58.7** | **54.2** |
| *Qwen3.5-9B* | | | | | | | | | | | | | | |
| ○ | Qwen3.5-9B | | 67.3 | 65.9 | 66.1 | 43.7 | 39.9 | 77.8 | 33.3 | 76.0 | 26.3 | 34.5 | 53.1 | 46.2 |
| ▲ | Kev-9B | joint | 62.2 | 62.3 | 60.5 | 37.7 | 36.5 | 76.8 | 31.4 | 56.0$^\ddagger$ | 23.7 | 36.1 | 48.3 | 45.5 |
| △ | Kev-9B | isolated | 63.3 | 64.0 | 60.1 | 38.7 | 34.8 | 75.9 | 33.3 | 71.7 | 21.9 | 35.6 | 49.9 | 44.5 |
| ▲ | Clef-Flash | joint | 67.5 | 67.4 | 63.9 | 42.9 | 41.2 | 80.9 | 34.1$^\ddagger$ | 62.4$^\ddagger$ | 20.0 | 37.5$^\ddagger$ | 51.8 | 46.1 |
| △ | Clef-Flash | isolated | 67.8 | 69.6$^\dagger$ | 64.5 | 43.8 | 43.1 | 80.5 | 36.1$^\dagger$ | 68.1$^\dagger$ | 23.4 | 40.8$^\dagger$ | 53.8 | 48.3 |
| ■ | Prism-9B | | **74.0**$^*$ | **71.1** | **71.0**$^*$ | 51.6$^*$ | 44.2 | 83.0 | **37.2** | **79.8**$^*$ | 34.7$^*$ | 43.2$^*$ | 59.0 | 53.7 |
| ■ | *Ours-9B* | | 73.8$^*$ | 70.0 | 67.3 | **52.1**$^*$ | **47.0** | **84.2** | 37.1 | 77.9$^*$ | **36.5**$^*$ | **47.7**$^*$ | **59.4** | **56.1** |
| *Qwen3.8-27B* | | | | | | | | | | | | | | |
| ○ | Qwen3.8-27B | | 71.9 | 68.7 | 66.9 | 45.8 | 44.7 | 82.6 | 37.2 | 74.5 | 30.1 | 41.7 | 56.4 | 51.5 |
| ▲ | Kev-27B | joint | 72.8 | **72.9**$^\dagger$ | **70.7**$^\dagger$ | 50.9$^\dagger$ | **49.2**$^\dagger$ | **83.9** | **38.4**$^\dagger$ | **81.1**$^\dagger$ | **36.1**$^\dagger$ | 44.9$^\dagger$ | **60.1** | **55.0** |
| △ | Kev-27B | isolated | 73.6 | 70.5 | 70.1 | 50.5 | 46.7 | 83.2 | 37.9 | 80.6 | 32.1 | 45.2 | 59.0 | 53.5 |
| ▲ | Clef | joint | 70.5 | 72.4 | 66.2 | 48.2 | 45.9 | 80.7 | 35.6 | 58.7 | 29.2 | 40.9 | 54.8 | 50.3 |
| △ | Clef | isolated | 69.4 | 68.5 | 66.5 | 43.7 | 43.3 | 80.3 | 36.2 | 62.7 | 28.1 | 40.5 | 53.9 | 49.6 |
| ■ | *Ours-27B* | | **74.6** | 69.6 | 69.0 | **52.3** | 48.3 | 83.0 | 38.0 | 79.2 | 34.6 | **47.1**$^*$ | 59.6 | 54.9 |

most three of ten collections ($^\dagger$). Kev-4B is even 2.5 points below its base model. Reranker fine-tuning of the same backbones adds 5.9 (Prism-9B) to 29.5 points (Qwen3-Reranker-0.6B). Put bluntly, a user who takes the released Qwen3.5-4B or Qwen3.5-9B and asks it whether a passage answers the query gets the same ranking as from the decision fine-tunes built on it, at no training cost.

*27B: the one place where decision fine-tuning helps.* The largest backbone is the exception. Kev-27B improves on its base model by 3.7 points (60.1 vs. 56.4), significantly on eight collections, and is the best system in our study. Clef, the other 27B decision fine-tune, does not improve on its base model (54.8 joint, 53.9 isolated). Our 27B reranker fine-tune, trained from the same checkpoint, gains about as much over the base model (59.6) and ranks as well as Kev-27B: 59.6 vs. 60.1 on average and 54.9 vs. 55.0 on the zero-shot subset. The two differ significantly on one collection only, DBPedia, where ours is better ($^*$). So at 27B, decision fine-tuning catches up with a plain reranker fine-tune; it does not overtake it. And Kev-27B is only 0.7 points ahead of Ours-9B, a reranker fine-tune a third of its size.

*Context.* Table 2 places the triplets among other rerankers on the same candidates. Jev (59.1) is as effective as the reranker fine-tunes of 8B to 9B (59.0 to 59.4) and the Set-Encoder (59.0), and slightly behind Kev-27B (60.1). The open Kev-27B thus matches the closed model it imitates, which suggests that there is no secret in Jev that an open fine-tune of a strong backbone cannot reach. Laya (43.5) does not improve on its BM25 candidates (43.2).

*Joint scoring helps only the largest models.* Scoring all candidates in one state gains about one point for Kev-27B (60.1 vs. 59.0) and Clef (54.8 vs. 53.9), but costs Kev-0.6B 12.2 points and Kev-8B 13.0 ($^\ddagger$; significant on all ten and on five collections). A likely reason is the length of the state. A joint state of 100 passages has about 13k tokens on TREC DL and 40k or more on BEIR, whereas Kev-0.8B and Kev-4B were trained on states of at most 7,552 tokens, and Kev-27B on states of up to 32,768 [21].

**Table 2: Context: other rerankers (nDCG@10, ×100) on the BM25 top-100 of TREC DL 2019–2023 and five BEIR collections; System One models in joint mode (Laya isolated), asked whether the passage answers the query (Sect. 4). †: published number on the same candidates [1, 35], not recomputed. Avg: mean over all ten collections (only where all are available). $/1k: mean API cost in US dollars per 1,000 queries over the collections run. Best reranker per column in bold.**

| | | TREC DL | | | | | BEIR | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Mode | DL19 | DL20 | DL21 | DL22 | DL23 | COVID | NFC | SciF | Touché | DBP | Avg | $/1k |
| BM25 | | 50.6 | 48.0 | 44.6 | 26.9 | 26.2 | 59.5 | 32.2 | 67.9 | 44.2 | 31.8 | 43.2 | – |
| *Listwise LLM rerankers* | | | | | | | | | | | | | |
| RankGPT-4 | | **75.6**† | 70.6† | – | – | – | **85.5**† | 38.5† | 75.0† | 38.6† | **47.1**† | – | – |
| RankZephyr-7B | | 74.3 | 70.0 | 67.9 | 50.5 | 44.1 | 80.7† | 36.6† | 75.1† | 31.1† | 43.2† | 57.3 | – |
| *Cross-encoders and LLM rerankers* | | | | | | | | | | | | | |
| bge-reranker-v2-m3 | | 72.2† | 67.0† | 66.1 | 51.5 | 45.1 | 74.8† | 33.8† | 73.5† | 39.9† | 41.9† | 56.6 | – |
| monoT5-3B | | 71.8 | 68.9 | 66.7 | 49.5 | 45.0 | 80.7† | **39.0**† | 76.6† | 32.4† | 44.5† | 57.5 | – |
| Qwen3-Reranker-8B | | 73.2 | 66.8 | 67.0 | 51.2 | 45.2 | 84.0 | 37.3 | 79.9 | **43.1** | 46.2 | 59.4 | – |
| Set-Encoder-large | | 75.0† | 72.2† | 69.9 | **53.4** | 47.7 | 81.2 | 36.8 | 73.8 | 33.2 | 47.0 | 59.0 | – |
| *System One models* | | | | | | | | | | | | | |
| Jev | joint | 73.0 | 70.9 | **70.7** | 50.3 | 46.1 | 83.3 | 37.8 | 80.3 | 33.5 | 45.5 | 59.1 | 1.11 |
| Clef-Flash | joint | 67.5 | 67.4 | 63.9 | 42.9 | 41.2 | 80.9 | 34.1 | 62.4 | 20.0 | 37.5 | 51.8 | – |
| Clef | joint | 70.5 | 72.4 | 66.2 | 48.2 | 45.9 | 80.7 | 35.6 | 58.7 | 29.2 | 40.9 | 54.8 | – |
| Kev-4B | joint | 62.5 | 63.2 | 60.8 | 37.0 | 36.3 | 75.3 | 29.3 | 58.4 | 19.9 | 39.8 | 48.2 | – |
| Kev-27B | joint | 72.8 | **72.9** | **70.7** | 50.9 | **49.2** | 83.9 | 38.4 | **81.1** | 36.1 | 44.9 | **60.1** | – |
| Laya | isolated | 59.5 | 54.6 | 52.8 | 35.3 | 32.7 | 73.8 | 28.7 | 44.8 | 21.8 | 31.3 | 43.5 | – |

**Table 3: Calibration and its uses on the judged pairs of TREC DL 2019–2023 (means). Per foundation model: base model, decision fine-tunes in their better-calibrated mode (j/i: joint/isolated), reranker fine-tunes. ECE (lower is better): N native; T, P after temperature/Platt scaling fitted on MS MARCO dev (rerankers only). Brier: mean squared error of the probability. $F_1$@.5 (×100): set kept at $p \geq 0.5$, native (N) and Platt-scaled (P); Regret: $F_1$ at each collection's best cutoff minus $F_1$@.5; $|S|$: passages kept at 0.5 (of 100). Count: mean absolute error of the summed probabilities against the number of relevant passages (always guessing the median count: 8.6). Per foundation model and column: best in bold, second best underlined.**

| | ECE | | | | $F_1$@.5 | | | | |
|---|---|---|---|---|---|---|---|---|---|
| System | N | T | P | Brier | N | P | Regret | $\|S\|$ | Count |
| *Qwen3-0.6B* | | | | | | | | | |
| ○ Qwen3-0.6B | 0.750 | – | – | 0.750 | 19.9 | – | **0.5** | 100.0 | 36.5 |
| △ Kev-0.6B (i) | **0.503** | – | – | **0.463** | 21.4 | – | 2.2 | 79.8 | **24.7** |
| ■ Qwen3-Reranker-0.6B | 0.511 | **0.326** | **0.226** | 0.481 | **31.8** | **6.3** | 14.9 | 58.4 | 24.8 |
| *Qwen3-8B* | | | | | | | | | |
| ○ Qwen3-8B | 0.231 | – | – | 0.233 | 42.5 | – | **2.1** | 22.9 | 8.7 |
| ▲ Kev-8B (j) | **0.113** | – | – | **0.177** | 34.0 | – | 3.1 | 13.7 | **7.6** |
| ■ Qwen3-Reranker-8B | 0.257 | **0.237** | **0.227** | 0.238 | **46.3** | **7.4** | 9.0 | 32.2 | 12.9 |
| *Qwen3.5-0.8B* | | | | | | | | | |
| ○ Qwen3.5-0.8B | 0.617 | – | – | 0.566 | 21.6 | – | 11.2 | 88.8 | 29.9 |
| △ Kev-0.8B (i) | 0.336 | – | – | 0.297 | 25.1 | – | 2.9 | 51.9 | 16.9 |
| ■ Prism-0.8B | **0.243** | **0.229** | **0.237** | **0.193** | **46.9** | **5.7** | **2.8** | 26.5 | **12.7** |
| *Other systems* | | | | | | | | | |
| ▲ Jev (j) | **0.075** | – | – | **0.127** | **54.7** | – | 1.7 | 16.0 | **6.3** |
| △ Laya | 0.127 | – | – | 0.183 | 37.5 | – | **1.0** | 23.3 | 9.2 |
| ■ monoT5-3B | 0.312 | 0.257 | **0.232** | 0.295 | 42.0 | **5.8** | 9.9 | 35.6 | 15.2 |
| ■ bge-reranker-v2-m3 | 0.231 | **0.226** | 0.236 | 0.228 | 48.2 | 5.7 | 3.8 | 29.3 | 12.1 |
| ■ Set-Encoder-large | 0.495 | 0.250 | 0.236 | 0.426 | 31.6 | 5.7 | 21.1 | 52.2 | 24.3 |

| | ECE | | | | $F_1$@.5 | | | | |
|---|---|---|---|---|---|---|---|---|---|
| System | N | T | P | Brier | N | P | Regret | $\|S\|$ | Count |
| *Qwen3.5-4B* | | | | | | | | | |
| ○ Qwen3.5-4B | 0.181 | – | – | 0.195 | 44.3 | – | 3.1 | 27.1 | 9.8 |
| △ Kev-4B (i) | **0.132** | – | – | 0.175 | 42.6 | – | **1.4** | 21.9 | **8.4** |
| ■ Prism-4B | 0.218 | **0.202** | **0.238** | **0.172** | **51.0** | **5.7** | 2.8 | 23.1 | 11.5 |
| *Qwen3.5-9B* | | | | | | | | | |
| ○ Qwen3.5-9B | 0.165 | – | – | 0.181 | 46.5 | – | 2.9 | 24.4 | 8.7 |
| ▲ Kev-9B (j) | 0.160 | – | – | 0.177 | 40.2 | – | 2.3 | 23.8 | 9.5 |
| ▲ Clef-Flash (j) | 0.112 | – | – | 0.158 | 47.1 | – | **1.7** | 18.9 | 8.1 |
| ■ Prism-9B | 0.242 | 0.233 | 0.238 | 0.183 | **50.7** | 5.7 | 4.0 | 26.0 | 12.3 |
| ■ *Ours-9B* | **0.111** | **0.108** | **0.231** | **0.157** | 46.6 | **6.9** | 6.4 | 8.7 | **7.3** |
| *Qwen3.8-27B* | | | | | | | | | |
| ○ Qwen3.8-27B | 0.117 | – | – | 0.144 | 54.1 | – | 1.9 | 16.0 | **5.8** |
| ▲ Kev-27B (j) | **0.076** | – | – | **0.130** | **57.2** | – | **0.9** | 16.1 | 6.0 |
| ▲ Clef (j) | 0.091 | – | – | 0.139 | 53.4 | – | 2.2 | 17.7 | 6.7 |
| ■ *Ours-27B* | 0.110 | **0.114** | **0.230** | 0.153 | 48.3 | **8.1** | 5.9 | 7.9 | 7.3 |

## 6 RQ2: Are Decision Models at Least Better Calibrated?

Calibration is the main promise of decision models. Table 3 reports it on the judged pairs of TREC DL 2019–2023, together with the two uses we test.

*Both kinds of fine-tuning improve on the small base models.* The zero-shot base models of 0.6B and 0.8B parameters are badly calibrated (ECE 0.750 and 0.617): they say "yes" to almost everything. Both decision and reranker fine-tuning fix much of this. From 4B on, however, the base models asked our question are already reasonably calibrated (ECE 0.117 to 0.231), and the gain from any fine-tuning is smaller.

*Decision fine-tunes beat the published rerankers, but not a plain reranker fine-tune.* With our question, the decision fine-tunes are better calibrated than the published reranker fine-tunes from 4B on (ECE: Kev-4B 0.132 vs. Prism-4B 0.218, Kev-8B 0.113 vs. Qwen3-Reranker-8B 0.257, Clef-Flash 0.112 vs. Prism-9B 0.242). At 0.6B they tie (0.503 vs. 0.511), and at 0.8B Prism is better (0.243 vs. 0.336). The best calibrated systems in the study are Jev (0.075) and the 27B decision fine-tunes (Kev-27B 0.076, Clef 0.091). This is the one result in favour of decision models, and it does not survive a closer look. First, our reranker fine-tune of Qwen3.5-9B, which asks no question

and was trained with a plain yes/no loss, is as well calibrated as the best decision fine-tune of its size (ECE 0.111 vs. 0.112) and has the lower Brier score (0.157 vs. 0.158). At 9B, calibration is a property of the training loss rather than of decision fine-tuning. At 27B the picture turns: Kev-27B is better calibrated than our reranker fine-tune (ECE 0.076 vs. 0.110, Brier 0.130 vs. 0.153). This is the one backbone where decision fine-tuning has a clear advantage in calibration over a reranker fine-tune of the same checkpoint. Second, the advantage over the published rerankers depends on the wording of the question: with S1Rank's question it is gone (Sect. 7).

*Calibration after the fact does not transfer.* Rerankers that are not calibrated can be fixed after training. Temperature scaling fitted on MS MARCO dev lowers the ECE of the worst calibrated rerankers on TREC DL (Qwen3-Reranker-0.6B from 0.511 to 0.326, the Set-Encoder from 0.495 to 0.250), but changes the others little. Platt scaling, which also shifts the scores, makes them worse at the fixed cutoff ($F_1$@.5 of 8.1 or less): MS MARCO dev judges about one passage per query, so the fitted probabilities are far too low for the denser judgments of TREC DL. Calibration is only defined with respect to a notion of relevance and its base rate. We return to this point in Sect. 8.

*Calibration does not make decision models select better passages.* If every passage with $p \geq 0.5$ is kept, a reranker fine-tune keeps the better set on every backbone up to 9B (e.g. $F_1$@.5 of 51.0 for Prism-4B vs. 42.6 for Kev-4B, and 50.7 for Prism-9B vs. 47.1 for Clef-Flash). At 27B, Kev-27B keeps the better set (57.2 vs. 48.3): our fine-tune is underconfident, keeps only 7.9 passages per query, and loses 5.9 points against its best thresholds. Calibration does help in one respect: for the decision models, 0.5 is close to the best threshold of each collection (regret of 0.9 to 3.1 points), whereas Prism-9B loses 4.0 points and Ours-9B 6.4 against their best thresholds. But the best threshold of a better ranker still yields the better set. A well-placed threshold on a worse ranking does not make up for the ranking.

*Counting relevant passages.* The count test asks for a use that ranking cannot serve: how many of a query's passages are relevant. Here most decision models from 4B on beat always guessing the median count (8.6), and they beat the published reranker fine-tunes of their backbones (Kev-8B 7.6 vs. Qwen3-Reranker-8B 12.9, Kev-4B 8.4 vs. Prism-4B 11.5). Decision fine-tuning is not what makes counting work, however. The untuned Qwen3.8-27B counts most accurately of all (5.8), ahead of Kev-27B (6.0) and Jev (6.3), and Ours-9B counts better than Clef-Flash (7.3 vs. 8.1).

## 7 Ablation: Does the Wording Matter?

Decision models are sold as models one can simply ask. So far we asked them one question, whether the passage answers the query. Table 4 repeats every run of a decision or base model with the question of S1Rank [27], which asks whether the passage is relevant and defines relevance more loosely: it "answers the query or contains information that directly addresses it". Both wordings are reasonable, and S1Rank's is the one under which Jev was first shown to be a good reranker.

*Ranking hardly moves.* The mean nDCG@10 of the decision fine-tunes changes by −2.5 to +2.6 points between the two questions, and our question is slightly worse for most of them (Jev joint 60.5 with S1Rank's question vs. 59.1 with ours, Kev-27B joint 61.1 vs. 60.1). None of the conclusions of RQ1 depends on the wording: the reranker fine-tunes ask no question, and with S1Rank's question the best decision fine-tune of every backbone up to 9B is still 4 points or more behind them.

*Calibration moves a lot.* The ECE, in contrast, drops for almost every model when it is asked our question: Jev joint from 0.224 to 0.075, Kev-8B joint from 0.243 to 0.113, Kev-27B joint from 0.156 to 0.076. The count error moves with it (Jev joint from 11.3 to 6.3 passages). With S1Rank's question, the decision fine-tunes up to 8B are no better calibrated than the published reranker fine-tunes of the same backbone (Kev-4B 0.216 vs. Prism-4B 0.218 in Table 3, Kev-8B 0.243 vs. Qwen3-Reranker-8B 0.257) or worse (Kev-0.8B 0.357 vs. Prism-0.8B 0.243). Clef-Flash keeps an edge over Prism-9B (0.140 vs. 0.242), but not over our 9B fine-tune (0.111). No decision model beats the median guess in the count test by more than a fraction of a passage (Clef joint 8.4 vs. 8.6). The answer to RQ2 thus flips with one sentence of the prompt.

*The base models move as much.* The untuned models gain as much from the wording as the decision fine-tunes (Qwen3-8B from 0.383 to 0.231, Qwen3.5-9B from 0.283 to 0.165). What matters is that the question matches the definition of relevance in the labels, not decision fine-tuning. A decision model's probability is calibrated against a notion of relevance that the user has to state in words, and a small change in the words changes the numbers by a factor of two or three. Reranker fine-tunes are not affected, since their notion of relevance is fixed by their training labels; they need no question to be worded, and none to be got wrong.

## 8 Discussion and Limitations

*Why decision fine-tuning does not make a reranker.* A reranker fine-tune is trained on the very task it is tested on: telling relevant from non-relevant passages of a query, with a ranking loss over hard negatives. The decision fine-tunes we tested learned to answer many kinds of questions about many kinds of states, and Kev's training data contain no MS MARCO and no BEIR [21]. That general skill transfers to relevance about as well as the base model's own instruction following, which is what we observe up to 9B. The zero-shot subset shows that the gap is not about having seen the test domain: it persists on collections that neither kind of fine-tune was trained on.

*What calibration means for relevance.* A probability of relevance is calibrated only with respect to a definition of relevance and its base rate. Our results show three ways to move it: the wording of the question (Sect. 7), the labels of the training data (our reranker fine-tunes) and the labels a scaling is fitted on (Platt scaling on MS MARCO dev, Sect. 6). A model that is "calibrated out of the box" is calibrated against a notion of relevance that someone chose, and the user of a decision model has to put that notion into words. For ranking none of this matters; for a threshold or a count it matters a lot.

**Table 4: The question ablation: S1Rank's question (*Rel.*: "Is document {doc} relevant to the query? Relevant means it answers the query or contains information that directly addresses it.") against the question used in the rest of the paper (*Ans.*: "Does document {doc} answer the query?"), for every system that is asked a question. nDCG@10 (×100): mean over all ten collections (Avg) and over TREC DL 2019–2023 (DL); ECE and Count as in Table 3 (DL 2019–2023). Δ: Ans. minus Rel. Training: ○ base model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune.**

| | | | nDCG@10 Avg | | | nDCG@10 DL | | | ECE | | | Count | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | System | Mode | Rel. | Ans. | Δ | Rel. | Ans. | Δ | Rel. | Ans. | Δ | Rel. | Ans. | Δ |
| *Qwen3-0.6B* | | | | | | | | | | | | | | |
| ○ | Qwen3-0.6B | | 23.8 | 26.6 | +2.8 | 25.4 | 30.7 | +5.3 | 0.750 | 0.750 | 0.000 | 36.5 | 36.5 | 0.0 |
| ▲ | Kev-0.6B | joint | 13.2 | 15.1 | +1.9 | 14.9 | 16.7 | +1.8 | 0.591 | 0.503 | −0.088 | 28.8 | 24.6 | −4.2 |
| △ | Kev-0.6B | isolated | 27.0 | 27.3 | +0.3 | 28.8 | 29.5 | +0.7 | 0.579 | 0.503 | −0.076 | 28.2 | 24.7 | −3.5 |
| *Qwen3-8B* | | | | | | | | | | | | | | |
| ○ | Qwen3-8B | | 53.3 | 51.1 | −2.2 | 56.9 | 54.0 | −2.9 | 0.383 | 0.231 | −0.152 | 17.5 | 8.7 | −8.8 |
| ▲ | Kev-8B | joint | 38.8 | 38.7 | −0.1 | 50.9 | 49.3 | −1.6 | 0.243 | 0.113 | −0.130 | 12.8 | 7.6 | −5.2 |
| △ | Kev-8B | isolated | 52.7 | 51.7 | −1.0 | 54.2 | 53.8 | −0.4 | 0.283 | 0.154 | −0.129 | 14.1 | 8.7 | −5.4 |
| *Qwen3.5-0.8B* | | | | | | | | | | | | | | |
| ○ | Qwen3.5-0.8B | | 37.6 | 33.9 | −3.7 | 40.3 | 39.1 | −1.2 | 0.716 | 0.617 | −0.099 | 34.8 | 29.9 | −4.9 |
| ▲ | Kev-0.8B | joint | 25.3 | 27.9 | +2.6 | 28.9 | 30.8 | +1.9 | 0.395 | 0.398 | +0.003 | 19.5 | 19.6 | +0.1 |
| △ | Kev-0.8B | isolated | 36.0 | 34.2 | −1.8 | 36.1 | 35.3 | −0.8 | 0.357 | 0.336 | −0.021 | 18.0 | 16.9 | −1.1 |
| *Qwen3.5-4B* | | | | | | | | | | | | | | |
| ○ | Qwen3.5-4B | | 52.7 | 52.4 | −0.3 | 55.8 | 56.1 | +0.3 | 0.296 | 0.181 | −0.115 | 14.7 | 9.8 | −4.9 |
| ▲ | Kev-4B | joint | 49.8 | 48.2 | −1.6 | 53.4 | 51.9 | −1.5 | 0.251 | 0.201 | −0.050 | 13.2 | 11.6 | −1.6 |
| △ | Kev-4B | isolated | 51.3 | 49.9 | −1.4 | 52.7 | 51.8 | −0.9 | 0.216 | 0.132 | −0.084 | 11.5 | 8.4 | −3.1 |
| *Qwen3.5-9B* | | | | | | | | | | | | | | |
| ○ | Qwen3.5-9B | | 53.7 | 53.1 | −0.6 | 56.7 | 56.6 | −0.1 | 0.283 | 0.165 | −0.118 | 14.1 | 8.7 | −5.4 |
| ▲ | Kev-9B | joint | 50.2 | 48.3 | −1.9 | 54.0 | 51.9 | −2.1 | 0.237 | 0.160 | −0.077 | 12.2 | 9.5 | −2.7 |
| △ | Kev-9B | isolated | 50.1 | 49.9 | −0.2 | 52.8 | 52.2 | −0.6 | 0.348 | 0.221 | −0.127 | 16.9 | 11.8 | −5.1 |
| ▲ | Clef-Flash | joint | 52.8 | 51.8 | −1.0 | 57.0 | 56.6 | −0.4 | 0.140 | 0.112 | −0.028 | 8.9 | 8.1 | −0.8 |
| △ | Clef-Flash | isolated | 54.7 | 53.8 | −0.9 | 58.9 | 57.8 | −1.1 | 0.173 | 0.144 | −0.029 | 9.7 | 8.2 | −1.5 |
| *Qwen3.8-27B* | | | | | | | | | | | | | | |
| ○ | Qwen3.8-27B | | 57.4 | 56.4 | −1.0 | 59.9 | 59.6 | −0.3 | 0.166 | 0.117 | −0.049 | 8.9 | 5.8 | −3.1 |
| ▲ | Kev-27B | joint | 61.1 | 60.1 | −1.0 | 63.5 | 63.3 | −0.2 | 0.156 | 0.076 | −0.080 | 8.5 | 6.0 | −2.5 |
| △ | Kev-27B | isolated | 59.8 | 59.0 | −0.8 | 62.5 | 62.3 | −0.2 | 0.262 | 0.180 | −0.082 | 12.9 | 9.4 | −3.5 |
| ▲ | Clef | joint | 56.1 | 54.8 | −1.3 | 61.1 | 60.6 | −0.5 | 0.147 | 0.091 | −0.056 | 8.4 | 6.7 | −1.7 |
| △ | Clef | isolated | 56.4 | 53.9 | −2.5 | 58.5 | 58.3 | −0.2 | 0.240 | 0.153 | −0.087 | 12.1 | 8.3 | −3.8 |
| *Other systems* | | | | | | | | | | | | | | |
| ▲ | Jev | joint | 60.5 | 59.1 | −1.4 | 62.7 | 62.2 | −0.5 | 0.224 | 0.075 | −0.149 | 11.3 | 6.3 | −5.0 |
| △ | Jev | isolated | 59.5 | 57.1 | −2.4 | 61.4 | 60.0 | −1.4 | 0.260 | 0.125 | −0.135 | 12.9 | 7.8 | −5.1 |
| △ | Laya | isolated | 44.0 | 43.5 | −0.5 | 47.3 | 47.0 | −0.3 | 0.131 | 0.127 | −0.004 | 9.3 | 9.2 | −0.1 |

*Limitations.* We compare released models, and the fine-tunes of a backbone differ in their training data as well as in their objective. Our results therefore describe decision fine-tuning as it is available today, not RLCD as such, which is unpublished and may be done better. The controlled comparison rests on our own reranker fine-tunes at 9B and 27B, which were trained once, with one recipe and without any hyperparameter search. We measured calibration only on TREC DL, where non-relevant passages are judged, and at a single relevance grade and a single threshold. Our question was fixed before the runs but not chosen on development data, and we tried only one other wording. All decision models are at most a month old, and newer versions may change the picture; we release all raw outputs so that the comparison can be repeated.

## 9 Conclusion

Do System One models make better rerankers? No. On the same backbone, plain reranker fine-tuning beats decision fine-tuning up to 9B parameters, by 5 to 29 points nDCG@10, and decision fine-tuning buys nothing over simply asking the base model. At 27B, where Kev is the best system in our study, a reranker fine-tune trained with a public recipe ranks as well, although Kev-27B is better calibrated. Calibration, the main promise of decision models, is largely a property of the question rather than of decision fine-tuning: when the question matches the relevance labels, even the base models become much better calibrated, and when it does not, the decision models lose their edge over reranker fine-tunes. Our reranker fine-tune, which asks no question, is as well calibrated as the best decision model of its size, and up to 9B no decision model kept better passages at a fixed threshold than the reranker fine-tune of its backbone. If you need a reranker, fine-tune a reranker.

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